NEXTGEN DEVSECOPS AI

Practical Generative AI for Beginners

Professional Training Curriculum & Learning Roadmap
Learn • Build • Experiment • Secure

2 MONTHS	2 HOURS / DAY	60–70 HOURS
Instructor-Led	Concepts + Labs	Approximate

GENAI	LOCAL AI	SECURE AI
LLMs, prompting and practical use cases.	Ollama, LangChain, ChromaDB and RAG.	DevOps, DevSecOps and AI Security awareness.

Format: Concepts + Live Demos + Labs + Assignments + Capstone

Prerequisites: Basic computer knowledge. No prior AI experience is required. Basic Python for LLM application development is taught as part of the program.

01 Course Overview

A beginner-friendly, practical journey into Generative AI. The curriculum progresses from essential development foundations to local LLMs, RAG and introductory AI Agents, with DevOps, DevSecOps and AI Security awareness built into the journey.

02 Learning Roadmap

Stage	Learning Path	Learner Gains
1	Git & GitHub Fundamentals	Version control and collaboration
2	Basic Python for LLMs	Programming foundation for GenAI
3	Linux & DevOps Foundations	Environments, CLI and Docker awareness
4	DevSecOps Awareness	Secure development and AI security mindset
5	Generative AI Fundamentals	LLMs, tokens, context and use cases
6	Prompt Engineering + Ollama	Prompting and local LLM execution
7	LangChain	Reusable LLM application workflows
8	ChromaDB + RAG	Grounded, document-aware AI applications
9	LangGraph + AI Agents	Stateful and multi-step agent workflows
10	Capstone Project	Integrated end-to-end GenAI solution

03 Module 1 — Generative AI Fundamentals

Module focus: Build a clear foundation without ML or Deep Learning prerequisites.

Topics & practical learning

- What Generative AI is and where it is used
- Large Language Models and how users interact with them
- Tokens, context windows, inference and model limitations
- Hallucinations, grounding and responsible use
- Common enterprise and productivity use cases
- Hands-on lab: explore LLM responses and identify strengths, limitations and suitable use cases.

04 Module 2 — Basic Python for LLMs

Module focus: Learn only the Python concepts needed for practical LLM applications.

Topics & practical learning

- Variables, data types, conditions and loops
- Functions and reusable code
- Lists, dictionaries and JSON
- Reading and writing files
- Modules, packages, pip and virtual environments
- Calling APIs from Python at a beginner level
- Hands-on lab: read input, process JSON and prepare data for an LLM workflow.

05 Module 3 — Git & GitHub Fundamentals

Module focus: Use professional version-control practices from the beginning.

Topics & practical learning

- Repositories, clone, add, commit, push and pull
- Branches and basic merge workflow
- README and project structure
- GitHub collaboration and pull-request awareness
- Basic .gitignore and safe repository practices
- Hands-on lab: create the course repository and manage changes through commits and branches.

06 Module 4 — Linux & DevOps Foundations

Module focus: Understand the environment and operational concepts supporting reliable AI applications.

Topics & practical learning

- Linux terminal navigation and file operations
- Permissions and environment variables
- Package and process basics
- Development environments and dependency consistency
- Docker concepts and why containers matter

- Build, test and deployment flow
- Hands-on lab: practice Linux commands and understand consistent AI application packaging.

07 Module 5 — DevSecOps & AI Security Awareness

Module focus: Introduce security at the right level for beginners.

Topics & practical learning

- Secure coding mindset
- Secrets and API-key protection
- Dependency and open-source risk awareness
- Prompt injection and unsafe-input awareness
- Sensitive-data and privacy considerations
- Responsible AI and basic secure AI development practices
- Hands-on lab: review exposed secrets, risky dependencies and prompt-injection scenarios.

08 Module 6 — Prompt Engineering

Module focus: Produce more consistent and useful LLM outputs.

Topics & practical learning

- Prompt anatomy: role, task, context and constraints
- Zero-shot and few-shot prompting
- Role-based and instruction prompting
- Structured outputs and formatting
- Prompt templates and reusable prompts
- Basic response evaluation and iterative improvement
- Hands-on lab: prompts for summarization, extraction, classification and structured output.

09 Module 7 — Ollama & Local LLMs

Module focus: Run open-source LLMs locally for practical experimentation.

Topics & practical learning

- Ollama installation and setup
- Downloading and running local models
- Model selection basics
- CLI interaction and model parameters
- Ollama REST API
- Connecting Python applications to a local LLM
- Hands-on lab: install Ollama, run a local model and invoke it from Python.

10 Module 8 — LangChain Fundamentals

Module focus: Move from direct model calls to reusable LLM workflows.

Topics & practical learning

- LangChain purpose and architecture at a beginner level
- Prompt templates
- Model invocation and output handling
- Output parsers
- Simple chains and reusable workflows
- Integration with Ollama
- Hands-on lab: build a simple LangChain application backed by a local LLM.

11 Module 9 — ChromaDB & RAG

Module focus: Build document-aware AI applications.

Topics & practical learning

- Why RAG is needed
- Document loading and text chunking
- Embeddings in the context of RAG

- Vector databases and ChromaDB
- Semantic retrieval
- Prompt + retrieved context flow
- Grounding and source-aware response concepts
- Hands-on lab: local document Q&A; / knowledge assistant using ChromaDB and Ollama.

12 Module 10 — LangGraph & AI Agents

Module focus: Introduce agentic patterns after prompts, LLM applications and RAG.

Topics & practical learning

- What an AI Agent is and when to use one
- Workflow vs agent concepts
- State and multi-step execution
- Nodes, edges and conditional flow in LangGraph
- Tool-use concepts
- Human-in-the-loop awareness
- Simple agent orchestration patterns
- Hands-on lab: build a small stateful LangGraph workflow.

13 Capstone Project

Build and demonstrate a secure local AI Assistant that integrates the technologies covered throughout the course.

Build	Integrate	Secure
Ollama + LangChain + ChromaDB/RAG	Git + Python + optional LangGraph	Basic DevOps, DevSecOps and AI Security practices

- Project planning and Git repository structure
- Local LLM integration with Ollama
- Prompt templates and LangChain workflow
- Document retrieval using ChromaDB and RAG
- Optional LangGraph agent workflow
- Basic DevOps and DevSecOps practices
- Testing, documentation and demonstration

14 Hands-on Lab Journey

Lab	Theme	Practical Outcome
Lab 1	Development setup	Git, Python, VS Code and course repository
Lab 2	Python for LLMs	Functions, JSON, files and virtual environments
Lab 3	Git workflow	Commit, branch, merge and clean repository practices
Lab 4	Linux & DevOps basics	Linux commands and environment/container concepts
Lab 5	Secure GenAI basics	Secrets, dependency risks and prompt-injection examples
Lab 6	Prompt Engineering	Zero-shot, few-shot, role and structured prompts
Lab 7	Local LLM with Ollama	Local model and Python API integration
Lab 8	LangChain application	Reusable prompt-to-model workflow
Lab 9	RAG with ChromaDB	Local knowledge assistant with learner documents
Lab 10	LangGraph Agent	Stateful multi-step workflow / simple agent
Capstone	Secure Local AI Assistant	Integrated GenAI solution with basic DevSecOps practices

15 Why DevOps & DevSecOps Matter for GenAI

- DevOps provides practical foundations for source code, environments, dependencies and deployment workflows.
- DevSecOps adds security practices such as protecting secrets, reviewing dependencies and considering security throughout development.
- Advanced CI/CD, Kubernetes and enterprise platform engineering remain outside the core beginner scope.

16 Learning Outcomes

- Understand Generative AI and LLM fundamentals without ML or Deep Learning knowledge.
- Use basic Python, Git/GitHub and Linux skills in a GenAI project.
- Run and integrate local LLMs with Ollama.
- Create effective prompts and reusable LangChain workflows.
- Build a document-based RAG application with ChromaDB.
- Understand and build introductory LangGraph / AI Agent workflows.
- Apply foundational DevOps, DevSecOps and AI Security awareness.
- Complete an end-to-end capstone suitable for demonstration.

17 Training Approach

Concept clarity → Live demonstrations → Guided hands-on labs → Assignments → Integrated capstone. The emphasis is on understanding how GenAI tools, Git, DevOps, DevSecOps and security practices fit into real-world AI application development.

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